

ADC PURE DC AI FACTORY

Eliminating Every Conversion Loss Between Fuel and Chip

A technical white paper on purpose-built, factory-tested 800V DC AI compute cassettes with liquid-first cooling, deployable onshore or offshore wherever DC power exists.

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1. EXECUTIVE SUMMARY

Every data center built today makes the same architectural compromise: it converts power from AC to DC and back again, multiple times, between the generator and the GPU. Each conversion stage wastes 2–5% of the energy passing through it. At the scale of modern AI training and inference workloads—thousands of GPUs running at hundreds of kilowatts per rack—those losses compound into megawatts of wasted power, millions of dollars in excess energy cost, and unnecessary carbon emissions.

The ADC Pure DC AI Factory eliminates this problem entirely. No AC exists anywhere in the system—not in generation, not in storage, not in distribution, and not in delivery to the chip. Power sources that generate DC natively (solid oxide fuel cells, solar panels, battery storage) feed an 800V DC bus that runs unbroken from the generation source to the NVIDIA GPU. The result is a purpose-built, factory-tested AI compute cassette that achieves a Power Usage Effectiveness (PUE) of 1.05 or better—approaching the theoretical minimum where virtually all input energy goes directly to computation.

Each cassette is a 40-foot ISO High Cube enclosure containing 10 NVIDIA NVL72 racks (Vera Rubin, Max P profile) delivering 2.3 MW of IT load with 720 GPUs, liquid-first cooling that removes over 80% of heat directly at the chip, and a complete Eaton power distribution stack—all factory-wired, factory-tested, and shipped ready for deployment. Site provides four connections: DC power, cooling loop, fiber, and a concrete pad. Time from pad-ready to first GPU workload: 30–60 days.

2.3 MW IT Load per Cassette	720 GPUs Vera Rubin per Cassette	≤ 1.05 Power Usage Effectiveness	800V DC End-to-End, No AC
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The cassette is deployable onshore on any prepared pad, or offshore on oil & gas platforms with existing natural gas generation—converting stranded energy into AI compute at the source. An offshore option package adds NEMA 4X / IP66 sealing, DNV 2.7-1 crane-lift certification, ISO 12944 C5-M marine corrosion protection, and hazardous area deployment strategies. The architecture is fuel- and grid-agnostic: any power source that delivers 800V DC at the external connection panel can run the cassette.

2. THE PROBLEM: CONVERSION LOSS AT SCALE

A conventional data center power chain looks like this:

Utility AC → Step-down Transformer → UPS (AC→DC→AC) → PDU → Server PSU (AC→DC) → VRM (DC→DC) → GPU

Each arrow represents a conversion stage. Transformers operate at 97–99% efficiency. UPS systems with double-conversion topology run at 92–96%. Server power supply units achieve 94–97% (80 PLUS Titanium). Voltage regulator modules on the board lose another 1–3%. Multiplied together, the power chain from utility feed to GPU die typically operates at 85–92% cumulative efficiency. The remaining 8–15% is dissipated as waste heat that the cooling system must then remove—consuming additional energy in the process.

At a single rack, this is a nuisance. At 62,640 GPUs across an 87-cassette deployment (the New Iberia reference site), it is a financial and engineering problem of first-order magnitude. A 5% improvement in power chain efficiency at 200 MW translates to 10 MW of recovered capacity—enough to power 43 additional racks of GPUs from the same fuel input, or equivalently, \$7–10 million per year in avoided energy cost at Louisiana industrial electricity rates.

The ADC Pure DC architecture eliminates conversion stages rather than optimizing them.

Bloom SOFC (DC native) → 800V DC Bus → Eaton ORV3 (±400V DC) → NVIDIA Kyber Power Shelf (54V DC) → GPU Point-of-Load

No transformer. No UPS. No AC anywhere. The fuel cell generates DC electrochemically. The battery stores and discharges DC. The solar panel outputs DC. The bus carries DC. The Eaton ORV3 steps DC voltage down. The NVIDIA Kyber shelf delivers DC to the GPU. Every component in the chain speaks the same electrical language, and every eliminated conversion stage recovers 2–5% of throughput power.

3. THE ARCHITECTURE: PURE DC, NO CONVERSION

3.1 The Cassette: Repeatable Compute Unit

The fundamental unit of the ADC system is the cassette: a purpose-built modular enclosure that contains a complete AI compute cluster. Unlike converted shipping containers or prefabricated data center modules that adapt AC infrastructure into a smaller form factor, the ADC cassette is designed from first principles for DC power delivery and liquid cooling. The enclosure is manufactured by Fibrebond in Minden, Louisiana—a company that has been engineering purpose-built modular structures for the data center and telecom industry for decades.

Each cassette uses a 40-foot by 8-foot by 9'6" High Cube ISO external form factor. This is not a converted shipping container; it is purpose-built steel using ISO dimensions for transport compatibility. The interior contains a single row of 10 NVIDIA NVL72 racks running the full length, a mechanical bay at the left end housing the cooling distribution unit, power distribution, environmental controls, and fire suppression, and an external connection panel at the right end where all site interfaces terminate. All utilities—DC power bus, cooling headers, network cabling—run overhead, leaving the rack row floor clear. Eight top-hinged access panels (four per side) provide NEC 110.26-compliant service access without requiring a walkable internal aisle.

3.2 Core Specification

Parameter	Value
Enclosure	40' × 8' × 9'6" HC ISO external — purpose-built steel
Racks	10 × NVIDIA NVL72 (OCP ORV3) — Vera Rubin Max P
GPUs per rack	72 × Vera Rubin (Rubin GPU + Vera CPU) — 288 GB HBM4/GPU — 20.7 TB HBM4 per rack
IT load per rack	230 kW
Total IT load	2,300 kW (2.3 MW)
Total facility draw	~2,415 kW — PUE ≤ 1.05
DC bus input	800V DC — OCP Stage 1d — 1500V insulation rated
DC bus current	~3,019A @ 800V — 3,774A conductor (NEC 125%)
Cooling	Boyd CDU 2,000 kW — N+1 pumps — >80% direct-to-chip
Layout	Single row — mech bay left — ext. panel right — utilities overhead
Access	8 top-hinged panels (4/side) — NEC 110.26 — no aisle required
Manufacturer	Fibrebond — Minden, Louisiana

4. POWER DISTRIBUTION: 800V DC END-TO-END

4.1 OCP Stage 1d Compliance

The cassette's power architecture is built on the Open Compute Project (OCP) Data Center Facility (DCF) Low-Voltage DC (LVDC) distribution framework, version 1.0 (March 2026). Key design constraints from this standard include a 1500 VDC ceiling per NEC and IEC, 800 VDC at the rack level, and TN-S or IT grounding only—no TN-C and no TT, which eliminates DC corrosion risk. The standard specifies solid-state circuit breakers (SSCBs) for microsecond DC fault response, which is critical because DC arcs do not self-extinguish at current zero-crossings the way AC arcs do. Every busbar, cable, and penetration in the cassette is specified against this OCP document.

4.2 Eaton ORV3 Distribution Stack

Power enters the cassette at the external connection panel as 800V DC. The PDU metering cabinet in the mechanical bay provides NIST-traceable power measurement at $\pm 1\%$ accuracy, main disconnect rated at 1,000A or greater per NEC 210.13(B), SSCB overcurrent protection with microsecond fault response, and ground fault protection relay with trip-confirmed reset. The 800V DC copper overhead busbar, sized at 3,774A nameplate (NEC 125% continuous load rule), runs the full length of the rack row and feeds three Eaton ORV3 sidecars.

ORV3 Sidecar	Serves	IT Load
ORV3-A	Racks R1–R4	920 kW — 800 kW rated — $\pm 400\text{V}$ DC to NVL72
ORV3-B	Racks R5–R8	920 kW — 800 kW rated — $\pm 400\text{V}$ DC to NVL72
ORV3-C	Racks R9–R10 + mech loads	460 kW IT + ~115 kW mech — 800 kW rated

Inside each NVL72 rack, the NVIDIA Kyber power shelf steps $\pm 400\text{V}$ DC down to 54V at the rack bus, then delivers point-of-load voltage at the GPU. Supercapacitor fast-cycle backup in the ORV3 sidecar provides sub-millisecond GPU load spike buffering and graceful shutdown holdup (≤ 500 ms) on power loss. The Eaton stack is co-designed with NVIDIA for DSX compliance.

Arc flash labeling per NFPA 70E is applied at all access points. Lockout/tagout capability is provided at the main disconnect. All grounding follows OCP DCF LVDC v1.0.

5. COOLING: LIQUID-FIRST AT PUE $\leq$ 1.05

5.1 Dual-Loop Architecture

Achieving PUE at or below 1.05 is not possible with air cooling at 230 kW per rack. Air-cooled facilities cannot sustain rack densities above approximately 100 kW without thermal throttling that reduces GPU boost clocks and effective compute throughput. The cassette uses a liquid-first architecture where direct liquid cooling at the GPU and HBM removes over 80% of rack heat before air is involved. The remaining residual heat (less than 20%) is managed through hot-aisle containment within the enclosure.

The cooling system operates as two independent loops:

Primary loop (inside cassette): A Boyd CDU rated at 2,000 kW liquid heat removal capacity circulates water-glycol coolant at approximately 1,300 liters per minute across all 10 racks (~130 LPM per rack). Cold supply enters at $\leq 45^{\circ}\text{C}$ and returns at $55\text{--}60^{\circ}\text{C}$. Each rack connects to the overhead supply and return headers via Staubli UQD hot-swap quick-disconnect fittings, enabling drip-free rack removal for maintenance without draining the loop. The CDU operates with N+1 internal pump configuration—either pump can carry the full cassette load alone. TraceTek rope-style leak detection runs the full loop per OCP Rope Leak Sensor Base Spec R1.0.0.

Secondary loop (site-provided, external): The CDU rejects heat to a site-provided dry cooler. The reference design specifies a BAC TrilliumSeries dry cooler with adiabatic assist, delivering approximately 1,840 kW of heat rejection capacity with zero evaporative water consumption under normal operating conditions. Adiabatic assist activates only during peak ambient days. The design day is 95°F at Lafayette, Louisiana (ASHRAE 0.4% design condition).

5.2 Zero Evaporative Water Consumption

Traditional data center cooling towers consume between 1.5 and 3 liters of water per kWh of IT load—a growing concern in water-stressed regions and an increasing regulatory constraint. The ADC closed-loop CDU plus dry cooler/adiabatic system eliminates evaporative water consumption during normal operations. The adiabatic assist mode, which uses minimal water during extreme heat events, is the only water consumption path, and it can be disabled entirely in water-sensitive deployments at the cost of slightly higher coolant supply temperatures.

Environmental control inside the cassette is handled by a Munters HCD or MCD series desiccant dehumidifier, installed in the mechanical bay. In Louisiana's 75–80% ambient relative humidity, dehumidification is essential before outside air contacts any electronics. The Munters unit regenerates using waste heat from the CDU return loop ($55\text{--}60^{\circ}\text{C}$), requiring no additional electrical draw for regeneration. Target interior humidity is $\leq 50\%$ RH per ASHRAE A1 guidelines (operating range: 20–80% non-condensing). The Munters supply creates positive pressure along the hot-aisle containment, ensuring conditioned air flows in the correct direction.

6. BUILDING MANAGEMENT & AUTONOMY

Each cassette is equipped with an NVIDIA Jetson AGX Orin (275 TOPS) edge AI controller functioning as the building management system (BMS). This is not a traditional PLC or SCADA controller—it is a purpose-built edge computing platform capable of real-time sensor aggregation, predictive analytics, and autonomous decision-making.

Function	Detail
Sensor aggregation	Temperature, humidity, leak, smoke, power anomaly — all channels read in real-time with $\pm 2\%$ calibration accuracy
Cooling PID control	CDU pump speed, coolant flow balance across racks, supply temperature setpoint management
CDU pump failover	Automatic N+1 pump switchover on primary pump fault — no operator intervention required
Alarm management	Temperature, humidity, leak, smoke, power anomaly → auto-escalate to site NOC via fiber link
Connectivity	Fiber to site NOC — local status display at external connection panel
Fire suppression	Honeywell VESDA-E VEU aspirating smoke detection with Ansul Sapphire Novec 1230 (FK-5-1-12) total-flood discharge

The cassette is designed to operate as an autonomous asset. Once commissioned and connected to its four site interfaces, it manages its own thermal environment, detects and responds to faults, and escalates to human operators only when intervention is required. This is critical for offshore deployments where on-site data center technicians may not be available.

Network Fabric

Cassette compute fabric is NVIDIA InfiniBand NDR at 400 Gb/s per port, native to the NVL72 rack. The spine switch is an NVIDIA QM9700 NDR, factory-mounted in the mechanical bay. All InfiniBand cables are factory-terminated and DSX compliance-validated before shipment. Cassette-to-cassette interconnect at scale uses site-provided passive copper (≤ 2 m) or active optical (> 2 m) links. External connectivity is 2× single-mode OS2 fiber at the external connection panel. The bay dimensions accommodate future upgrade to Quantum-X800 XDR form factor.

7. OFFSHORE & HARSH ENVIRONMENT DEPLOYMENT

The cassette's pure DC architecture and self-contained design make it uniquely suited for deployment on offshore oil & gas platforms and remote industrial facilities. These sites have abundant natural gas (often flared or vented as waste), existing power generation infrastructure, and an acute need for edge AI compute for seismic processing, predictive maintenance, autonomous operations, and real-time production optimization. An offshore option package extends the baseline onshore cassette with the following environmental hardening—all implemented as a priced delta, leaving the core architecture unchanged.

NEMA 4X / IP66 Enclosure Sealing	ISO 12944 C5-M Corrosion Class	DNV 2.7-1 Lift Certification	≤ 20 MT Crane-Lift Target
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7.1 Enclosure & Environmental Sealing

The baseline onshore cassette is rated NEMA 3R (rain, sleet, external ice formation). The offshore package upgrades to NEMA 4X per NEMA 250, dual-rated to IEC 60529 IP66 for international classification society acceptance. All external panels use 316L stainless steel or marine-grade fiberglass composite. Gaskets and seals are silicone or EPDM with UV stabilizer rated for 20-year Gulf of Mexico exposure. Every bulkhead penetration—Staubli UQD, DC bus entry, coolant, fiber, BMS, suppression—is sealed via IP66-rated cable glands or marine-rated connectors. Salt spray qualification per ASTM B117 at 1,000 hours minimum. The Munters desiccant unit is upsized for 95°F / 95% RH worst-case marine ambient with continuous positive internal pressure to prevent salt-laden air ingress.

7.2 Structural & DNV 2.7-1 Certification

An offshore cassette is not just a container—it is a marine-rated lifting module. The structural frame is certified to DNV 2.7-1 (Offshore Containers) covering dynamic crane lift in sea states up to SS4, pad eye certification with NDE-inspected welds, sling angle calculations, and proof load testing. ISO 1161 twist-lock corner castings are rated for dynamic load factors per DNV 2.7-1 §5.2. A floating internal subframe isolates the rack row and CDU from hull-transmitted vibration, with resilient mounts rated for 5–50 Hz continuous excitation from platform rotating machinery. The target deployed weight (loaded with empty racks, CDU with coolant, all mechanical) is under 17 metric tons—well within the 20 MT crane envelope common on modern platforms. GPU trays are installed on-platform after cassette placement.

7.3 Hazardous Area Strategy

None of the IT equipment inside the cassette is Ex-rated. Two deployment strategies apply. **Strategy A (preferred):** locate the cassette in an unclassified zone of the platform—living quarters deck, dedicated safe module area, or purpose-built equipment platform outside the classified zone boundary. No Ex modifications required. **Strategy B:** for Zone 2 (IEC 60079) or Division 2 (NEC) placement, upgrade the enclosure to IEC 60079-2 pressurized enclosure (Ex p). The Munters desiccant air handling unit serves dual duty as the purge/pressurization source, maintaining continuous positive internal pressure ≥0.5 mbar above ambient with a pre-purge cycle before energizing IT load and automatic de-energization on pressure loss.

7.4 Platform Power Interface

Most offshore platforms generate power from gas turbines producing 3-phase AC at 4.16 kV or 13.8 kV. A separate 20-foot AC-to-DC rectifier module bridges the platform power bus to the cassette's 800V DC input. This keeps the cassette design pure while providing a turnkey power interface. Where platforms flare excess associated gas, a dedicated gas genset (e.g., Caterpillar CG260, Waukesha APG) can be sized for the 2.5 MW cassette load with waste heat available for Munters regeneration—converting gas that would otherwise burn unproductively into AI compute.

8. MANUFACTURING & SUPPLY CHAIN

The ADC cassette is manufactured entirely in the United States, with a Louisiana-anchored supply chain designed for production scale, not prototype builds.

Partner	Role & Location
Fibrebond	Enclosure manufacturing, factory integration, and testing — Minden, Louisiana
Eaton	ORV3 sidecars, PDU, SSCB, main disconnect, Boyd CDU (Eaton-acquired) — factory-installed by Fibrebond
NVIDIA	NVL72 racks, Vera Rubin GPUs, Kyber power shelves, QM9700 spine switch, InfiniBand NDR fabric, Jetson AGX Orin BMS
Bloom Energy	SOFC fuel cells — DC-native power generation — 90-day delivery
First Solar	Series 7 thin-film solar panels — New Iberia, 30 mi from reference site
Munters	HCD/MCD desiccant dehumidifier — waste-heat regenerated
BAC	TrilliumSeries dry cooler / adiabatic assist — site-provided
Staubli	UQD hot-swap quick-disconnect fittings — Boyd-stocked, NVIDIA-approved
Ansul	Sapphire Novec 1230 fire suppression — NFPA 2001 total flood

Every cassette undergoes a Factory Acceptance Test (FAT) before leaving Minden. The FAT protocol includes hydrostatic cooling loop pressure test (1.5× operating, 30-minute hold), 1500V DC insulation resistance (Megger test per IEEE 43), ground fault protection verification, BMS sensor calibration against NIST-traceable references, VESDA functional test, leak detection continuity, Staubli UQD connect/disconnect and flow test, fire suppression inspection, all access panel cycling, network fabric link-up, and power meter calibration. A signed FAT report ships with every cassette.

The cassette is TAA (Trade Agreements Act) compliant and US-manufactured throughout. This qualification is increasingly important for federal, defense, and critical infrastructure procurement.

9. SCALE: FROM PROOF OF CONCEPT TO 200 MW

The cassette is the repeatable unit. Scale is achieved by deploying more cassettes on more pads, connected to more DC power sources. The architecture does not change between a 4-cassette proof of concept and an 87-cassette production site—only the number of units and the site infrastructure grow.

Site	Cassettes / Racks / GPUs	IT Load
MARLIE 1 (proof of concept)	4 cassettes — 40 racks — 2,880 GPUs	9.2 MW
Trappeys Phase 1	5 cassettes — 50 racks — 3,600 GPUs	11.5 MW
Trappeys Full Build	50 cassettes — 500 racks — 36,000 GPUs	115 MW
New Iberia	87 cassettes — 870 racks — 62,640 GPUs	200 MW

Each site requires the same infrastructure pattern: a cassette pad array with spacing, orientation, crane access lanes, and service clearances; a DC bus infrastructure routing DC power from the generation zone to cassette tie-in points; a secondary cooling loop from CDU to site-level dry coolers; BESS placement on a separate pad per FM Global DS 5-33; a generation zone with Bloom SOFC units (N+1) and gas line routing; and a NOC/operations building with fiber hub, BMS aggregation, and technician workspace.

Time from pad-ready to first GPU workload is 30–60 days per cassette, compared to 18–24 months for traditional purpose-built data center facilities. This compression is possible because the cassette arrives factory-tested with all internal systems commissioned. Site work is limited to the four external connections and pad preparation. Each week between pad-ready and first workload is lost revenue at 720 GPUs per cassette.

Performance Comparison: ADC vs. Traditional

Metric	ADC Cassette	Traditional Air-Cooled
PUE	≤ 1.05	1.3–1.5
Liquid heat removal	>80% direct-to-chip	0% liquid path
GPU thermal throttle	None — full TDP sustained	Common above ~100 kW/rack
Time to compute	30–60 days pad-ready to live	18–24 months traditional build
Cooling redundancy	N+1 pumps, hot-swap capable	No planned cooling downtime
Power source	Any 800V DC source	Grid and/or genset (AC)
Manufacturing	Factory-tested, signed FAT report	Field integration risk
Water consumption	Zero (normal ops)	1.5–3 L/kWh (cooling towers)

10. CONCLUSION

The ADC Pure DC AI Factory represents a fundamental rethinking of how AI compute infrastructure is built, powered, cooled, and deployed. Rather than optimizing the legacy AC power chain—adding incremental efficiency points to transformers, UPS systems, and power supplies—the architecture eliminates the conversion stages entirely. The result is a system where virtually all input energy reaches the GPU, heat is removed directly at the chip before air is involved, and the complete compute cluster ships factory-tested in a single liftable module.

The pure DC architecture is not just more efficient—it is more deployable. Because the cassette accepts any power source that delivers 800V DC, it is agnostic to the energy generation method. Bloom SOFC fuel cells running on natural gas, First Solar panels, battery energy storage, or a rectified utility feed all connect to the same external panel. This fuel agnosticism is what makes offshore deployment viable: wherever there is stranded natural gas and a flat surface, the cassette can convert waste energy into AI compute.

The Louisiana supply chain—Fibrebond for manufacturing, Eaton for the complete power and cooling stack, Bloom Energy for DC-native generation, First Solar for onsite solar, all within a 200-mile radius of the reference sites—provides production capacity, not prototype assembly. The cassette is a manufactured product with a factory acceptance test protocol, not a custom engineering project.

The DC bus is the moat. Every eliminated conversion stage is 2–5% efficiency recovered—compounding across tens of thousands of GPUs. At 200 MW, that recovered efficiency is the difference between 62,640 GPUs running at full throttle and thousands of GPUs worth of energy lost to heat before it ever reaches a chip.

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